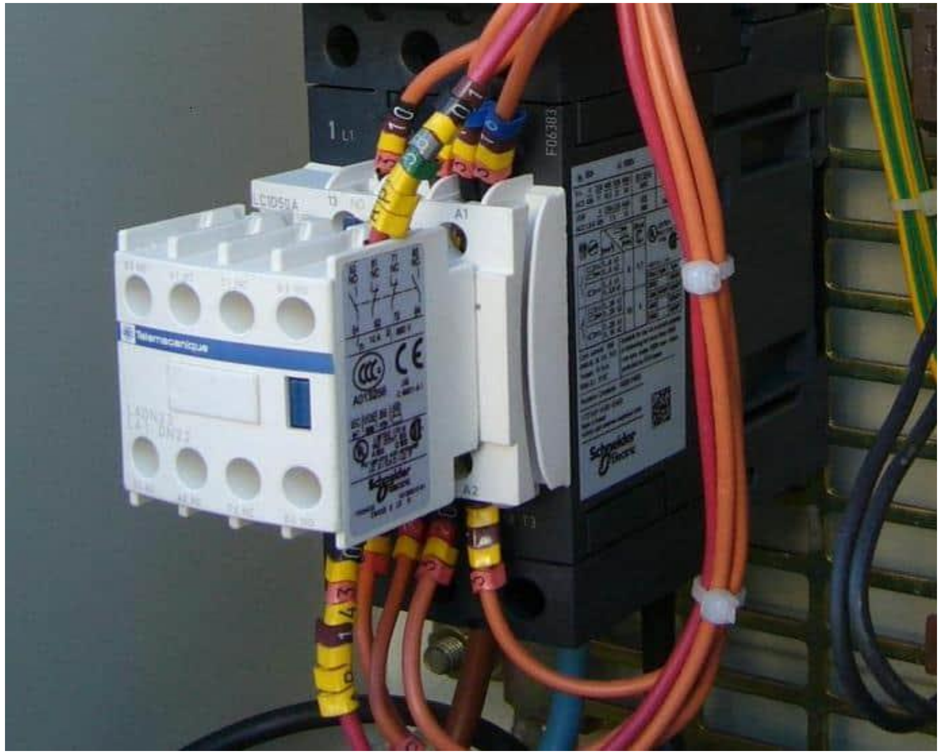
# Electrical Contactor

Lecture 13

# Electrical Contactor

- A contactor is an electromechanical control device that is used to make or break the connection between the load and the power supply. The use of a contactor is similar to the relay. But the device used for higher current-carrying applications is known as a contactor, and the device used for lower current applications is known as a Relay.
- A contactor has several contacts as per the application and load. Generally, these contacts are normally open (NO) contacts. And hence the load is shut off when the coil of the contactor is de-energized. But the contactor can be designed for both normally open and normally closed applications. The most common application of a contactor is in the starter that is used to turn ON and OFF the equipment, like motor, transformer, etc.

A contactor is an electrically controlled switch used for switching a power circuit, similar to a relay except with higher current ratings. A contactor is controlled by a circuit that has a much lower power level than the switched circuit. Contactors are often used for a 150 Hp motor.



# Construction of Contactor

- A contactor has three main parts;
- Coil or Electromagnet
- Encloser or Frame
- Contacts

# Coil or Electromagnet

- The coil is wound on an electromagnetic core and behaves as an electromagnet. Generally, it has two parts, one is a fixed part and the second is a movable part. A spring is connected between both parts. Hence, there is a spring return arrangement. A rod is connected with the moving part. This rode is also known as an armature. When a force of coil is more than the force of spring, both contacts are connected and when the force of spring is more than the force of the coil, both contacts are extracted with each other.
- A very small amount of current will flow through the spring from the supply or external control circuit to excite the core of the electromagnet.

- For AC applications, the electromagnetic core is made up of laminated soft iron to reduce the eddy current. For DC applications, there is no issue of eddy current, the core is made up of solid steel.

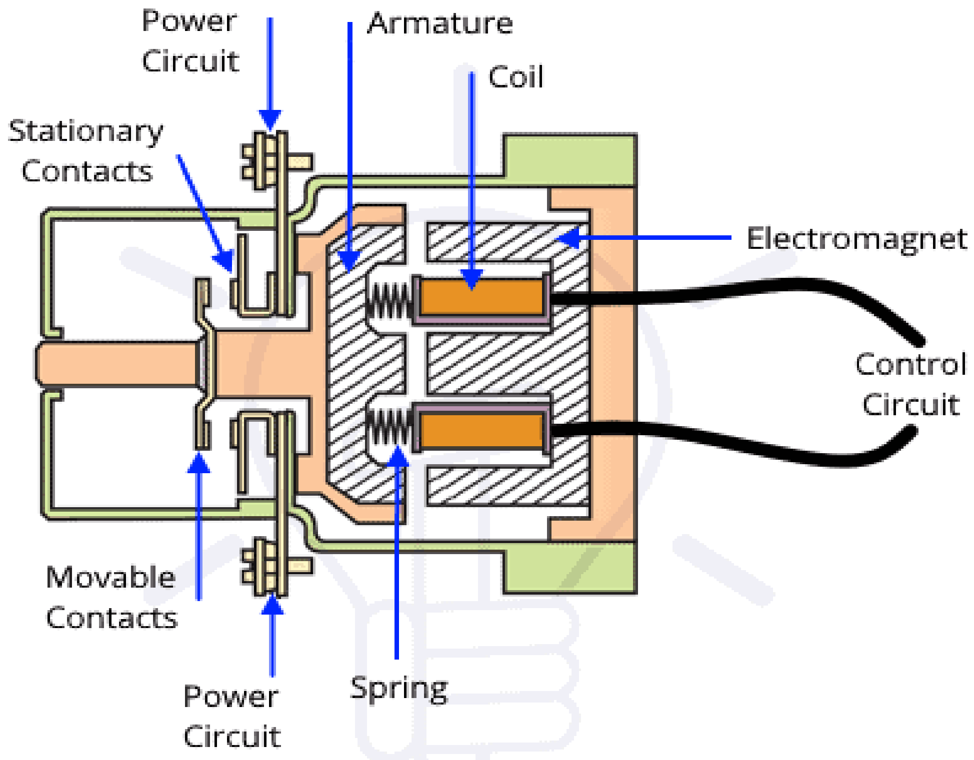
# Enclosure or Frame

- The Enclosure is used to protect the internal parts of the contactor. It is made up of plastic, nylon 6, ceramic, or Bakelite.
- It provides housing for the electromagnet and contacts.
- The enclosure is used to insulate the contacts and provide protection from the dust, oil, weather, and other explosion hazards. It avoids direct touching of contact when it is powered.

# Contacts

- This is the only component from which the entire load current will flow. Hence, it is a very important component of the contactor. The contacts are classified as power contact, auxiliary contact, and contact spring. There are two types of power contact; stationary contact and movable contact.
- The material used for the contacts has stable arc resistance and high welding resistance. These materials must withstand mechanical stress, erosion, and arc. The resistance of this material is as low as possible because the full load current will pass through the contacts.
- For the low current application, these contacts are made up of silver cadmium oxide and silver nickel, and for high current application and DC current, it is made up of silver tin oxide.

The armature of the electromagnet is connected with the moving contact. Hence, the moving contact moves with the action of an electromagnet and connect/disconnect with the fixed contact.




CONSTRUCTION OF CONTACTOR

# Working of a Contactor

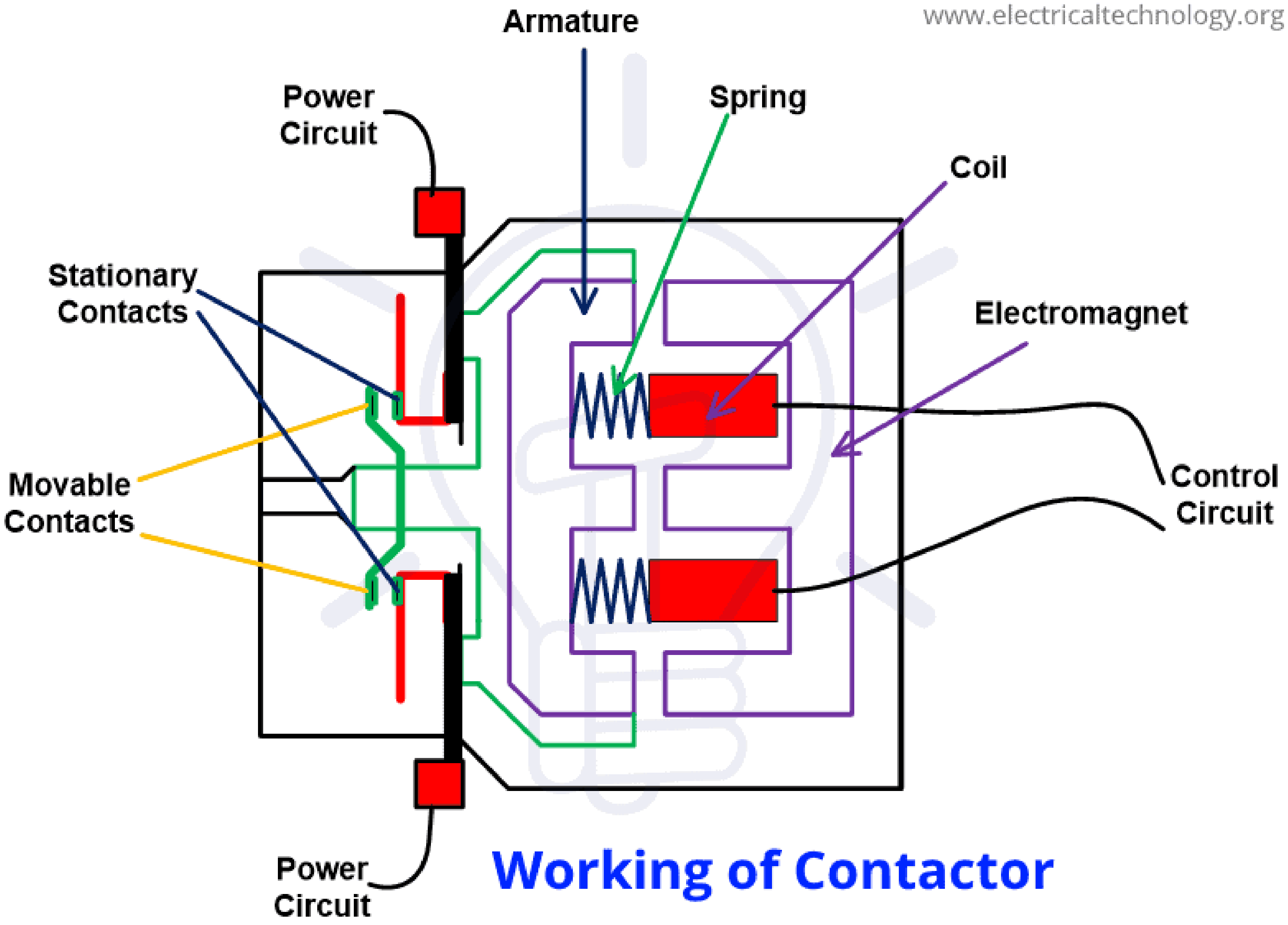
- An electromagnetic field is produced when the electromagnetic coil is energized. The moving contact of the contactor is connected with the armature (metallic rod) of an electromagnet.
- When an electromagnetic field is produced, the armature experiences the force and pulls towards the fixed contact. The force produced by the coil is more than the force of the spring. Both contacts remain in this position until the coil is not de-energized.

Once the coil is de-energized, the electromagnetic force is zero, and the armature pulls back due to the force of the spring. And return to the normal condition (OFF position). The contactors are designed for the rapid ON-OFF operation.

The input of the contactor coil may be AC or DC, or in some cases, the universal coil is used as an electromagnetic coil. The universal coils operate on both AC and DC.

A small amount of power loss occurs in the contacts, and an economizer circuit is used to reduce this loss.

While making and breaking contacts, an arc is produced between the contacts. This arc may reduce the life of the contactor as it increases the temperature of the contacts. Due to the arc, harmful gases are produced, like mono-oxide. Hence, there are several methods used to control and extinguish arcs.



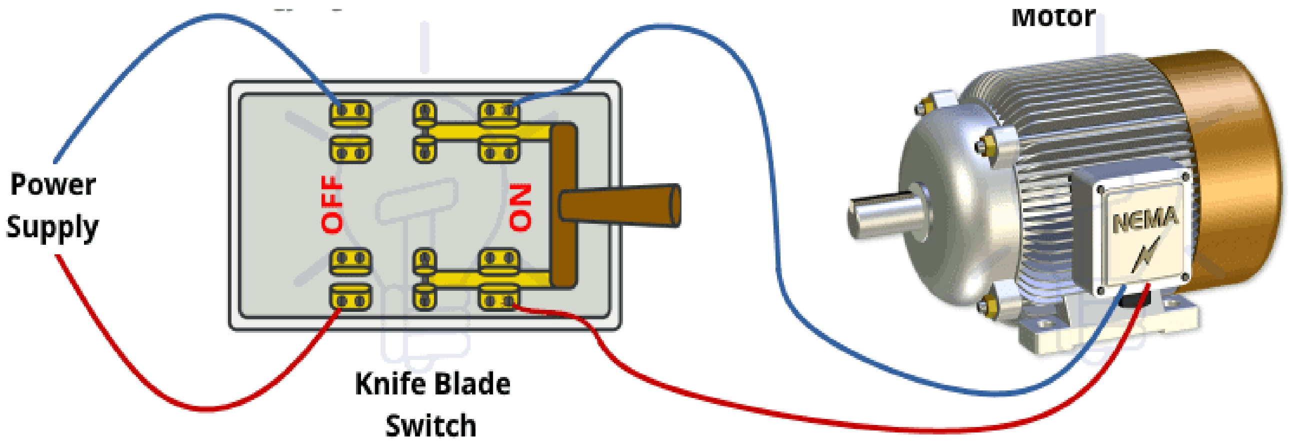

Working of Contactor

The contactors are selected base on load current and voltage, the control range of voltage, and application based on utilization category. If you want to check the connections of contacts are open or closed, you can do it with the help of an ohm-meter. Connect the ohmmeter between the input and output contacts. If the meter shows an infinite reading, the contacts are open, and if the meter shows a zero reading, the contacts are closed.

# Types of Contactors

- *Knife Blade Switch*
- This is the oldest type of contactor that used ON and OFF electric motors. It consists of a metal strip with a lever. The lever is used to pull-up and pull-down metal strips. This is a manual contactor. And it is very difficult to fast turn ON and OFF manually. There is also a chance to wear out the contacts.
- The full load current will pass through the contacts, and hence for a large motor, the full load current is very high. In this condition, there is a problem of arc production between the contacts, and it is difficult to quench the arc.

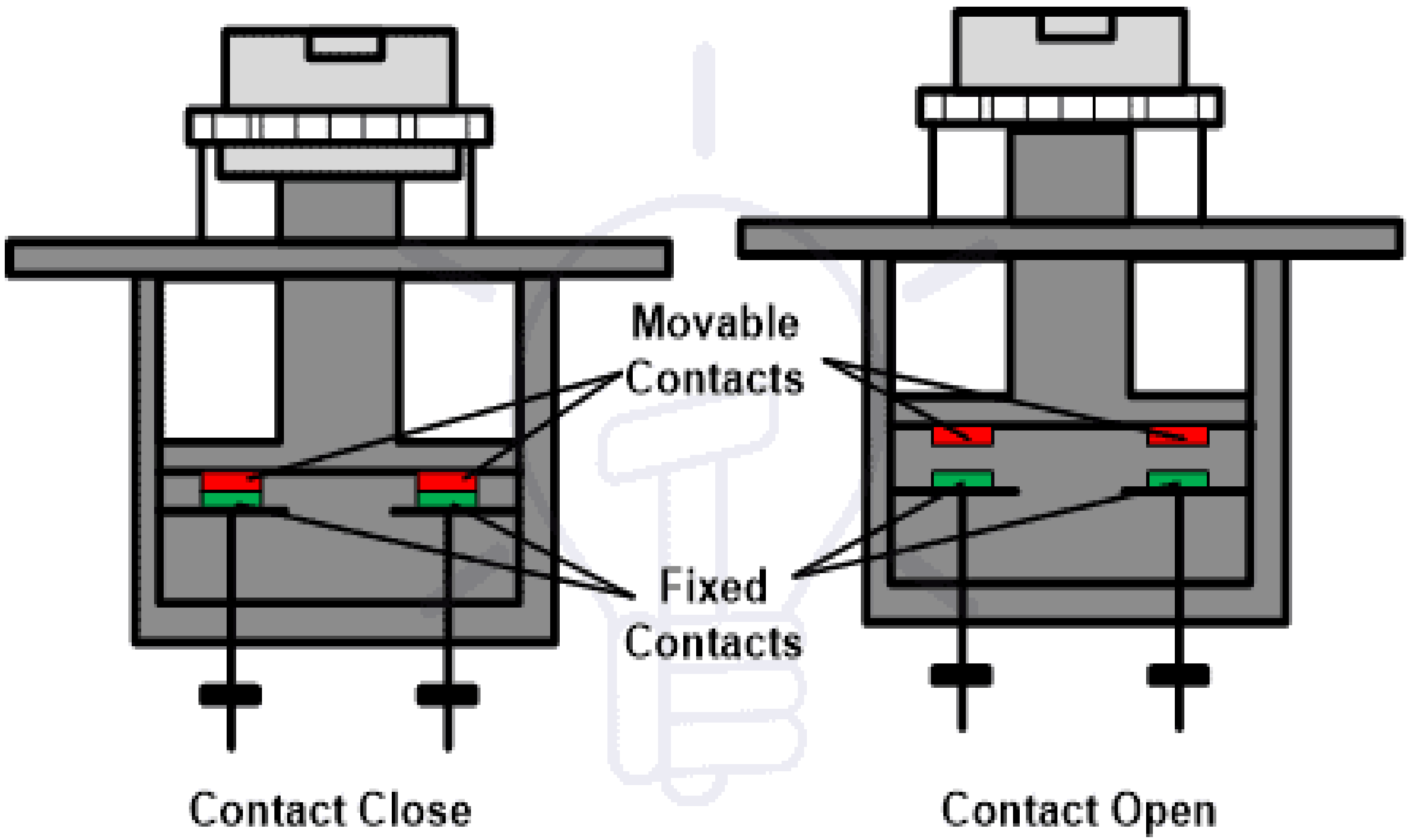
The second problem is the power loss.
As the current is very high, a large amount of power will drain by the contacts.
And the third problem is safety. Therefore, this type of contactor needs improvement. The life of this contactor is very short as there is a chance of corrosion of contacts due to moisture. Due to the problems and risks of the operation, this contactor is rarely used.




Wiring & Working of Knife Blade Switch

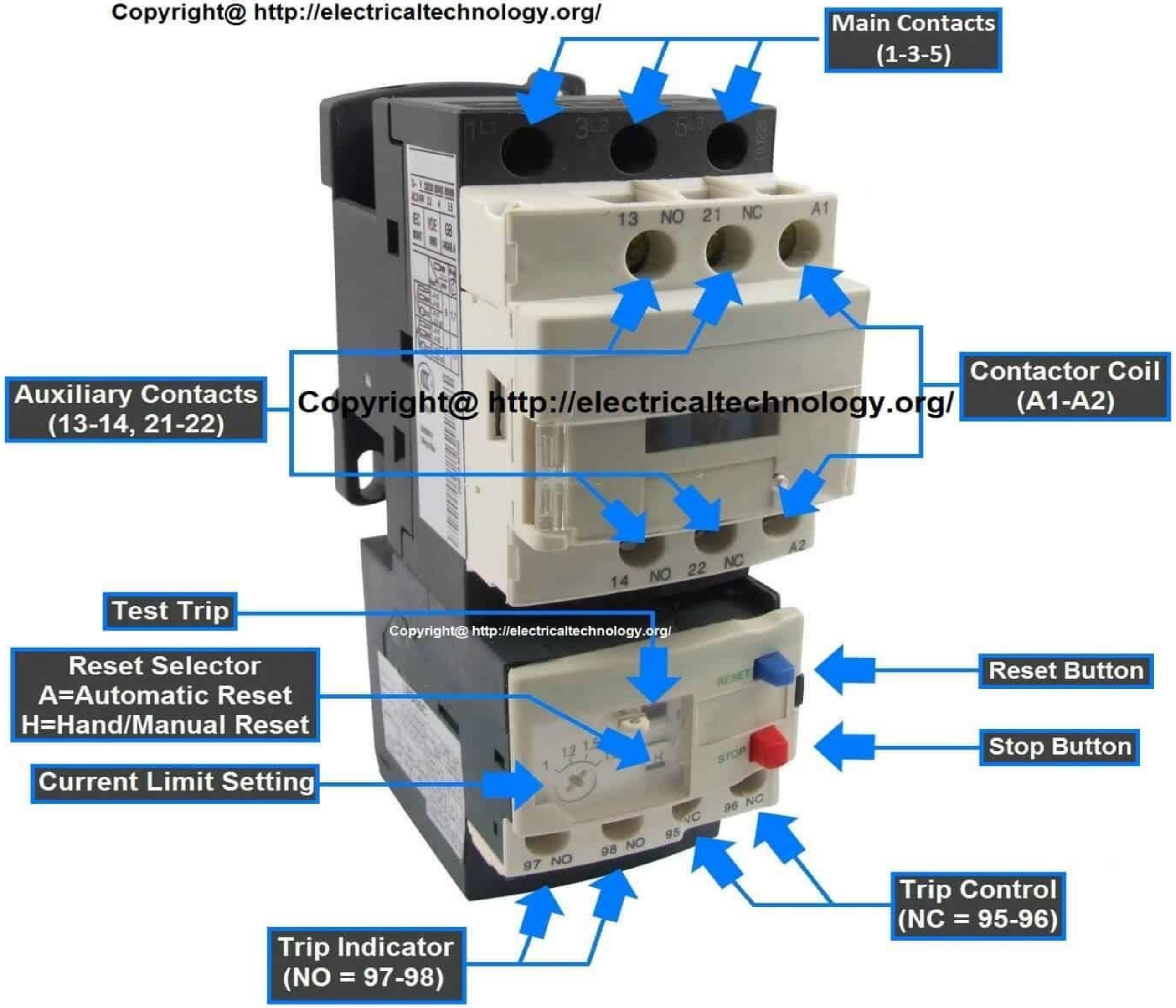
## Manual Contactor (Double Break Contactor)

There are lots of disadvantages to knife blade contactors. Hence, to overcome these disadvantages, a manual contactor was invented. This type of contactor is safe to operate with a smaller unit.



## Magnetic Contactor

The design of this type of contactor is the most advanced among all other types of contactors. This is an electromagnetic type contactor, and it can operate automatically. It requires a small amount of control circuit to turn on and off the load. Therefore, the operation of this contactor is safe compared to manual contactor. This is the most used contactor in industrial applications. It operates electromechanically and hence; it requires a very small amount of current to make a connection between the load and power supply.




# Magnetic Contactor

# Application of Contactors

The contactor is used in the following applications.

- The most common application of contactor is in the motor starter. It is used with overload and short circuit protection for the industrial motor.

- The contactors are used for the automation of lights for industrial, commercial, and residential lighting applications. For this type of application, latch type relay is used. In this type of relay, two coils are used. One for open contact and second for close contact.

- Single-pole contactors are used to operate 12 VDC load in the vehicle.

- The use of contactors with the circuit breaker assures the safety of the operation of load in industries. And in such an application, it is used to fast switching of a load.

- It is used in mercury relay and mercury-wetted relay.

- Two-pole (3-wire, 1-phase) contactors are used to operate 240 VAC load, like an air conditioner.

# The difference between contactor and relay

- The contactor is used for high voltage switching applications, and the relay is used for low voltage switching applications. Generally, if the load current is more than 15A, the contactors are used and if the load current is less then 15A, the relay is used.
- The size of the contactor is large compared to the size of the relay.
- The maintenance of contactor is easy while in most of the conditions, the relay cannot repair.
- In most cases, the contactors are connected in normally open contacts and the relays are connected as normally closed contacts.
- The switching time of the contactor is slow compared to the relay.